14 The ratio $(y + x) : (y - x)$ is equivalent to $k : 1$

Show that $y = \dfrac{x(k+1)}{k-1}$

Equal ratios give $y + x = k(y - x)$.

1 mark
Form an equation from the ratio

×	y	$-x$
k	ky	$-kx$

Expand: $y + x = ky - kx$

Add kx to both sides, then subtract y from both sides:

$x + kx = ky - y$

1 mark
Separate the x and y terms

×	1	k
x	x	kx

×	k	-1
y	ky	$-y$

So $x(k + 1) = y(k - 1)$. Divide by $k - 1$, with $k \neq 1$.

$y = \dfrac{x(k+1)}{k-1}$

1 mark
Factorise and complete the rearrangement

(Total for Question 14 is 3 marks)

15 $x = 0.4\dot{3}\dot{6}$

Prove algebraically that x can be written as $\dfrac{24}{55}$

(Total for Question 15 is 3 marks)

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